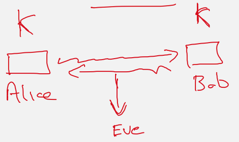
Symmetric key cryptography methods

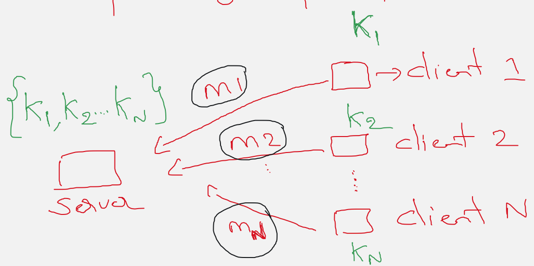

K
K
Alice
Bob
Eve

1) DES
2) AES

① Key distribution?

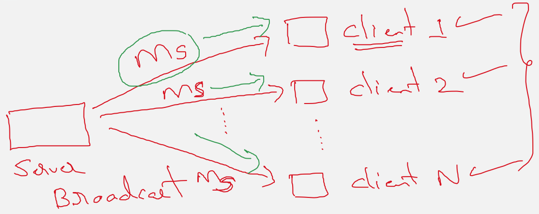
public key cryptography
↳ RSA
↳ ECC

Basic difference between Symmetric key and public key cryptography

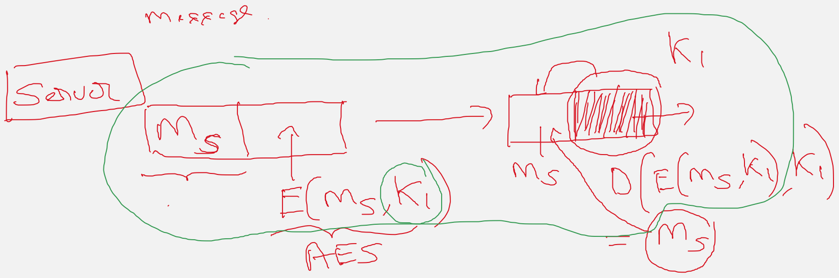

K_1
$\{K_1, K_2 \dots K_N\}$
m_1
client 1
K_2
m_2
client 2
server
m_N
client N
K_N

Scalability N is large
↳ Key distribution
↳ Key management at the server

Authentication


m_s
client 1
m_s
client 2
server
Broadcast m_s
client N

client i needs to authenticate the source message?


Server
m_s
K_1
$E(m_s, K_1)$
AES
m_s
$D(E(m_s, K_1), K_1)$
$= m_s$

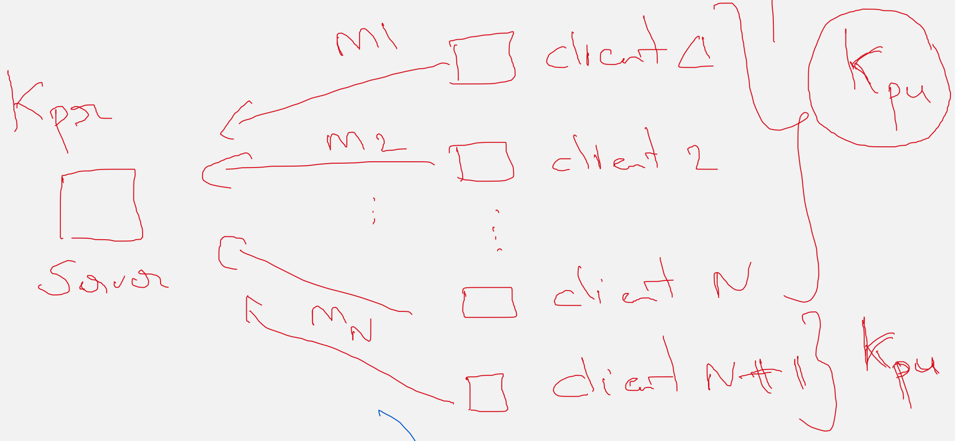
AES/DES are NOT suitable for multiparty transactions.

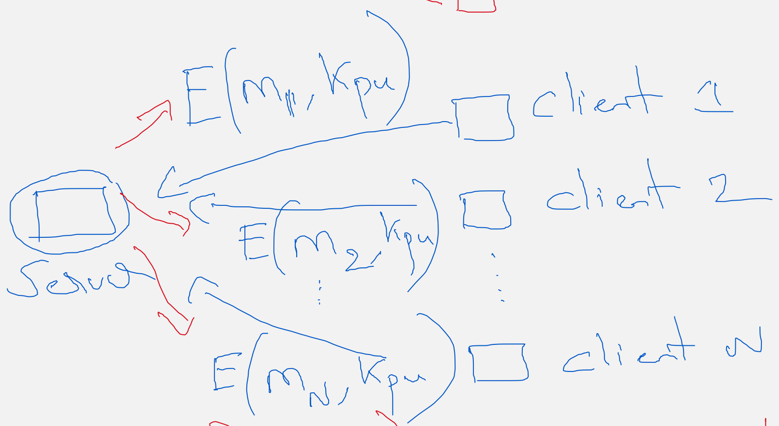
fundamental concepts

1) Key generation phase K_{pr} K_{pu}
2) Encryption
3) Decryption

K_{pr} → known only to the server (recipient)

K_{pu} → Know to everybody
→ public


K_pr
Server
m1
m2
m_N
client 1
client 2
client N
client N+1
K_pu
K_pu


E(m_1, K_pu)
E(m_2, K_pu)
E(m_N, K_pu)
Server
client 1
client 2
client N

$$D(\underbrace{E(m_1, K_{pu})}, K_{pr}) \Rightarrow m_1 \checkmark$$

$$D(E(m_2, K_{pu}), K_{pr}) \Rightarrow m_2 \ -$$

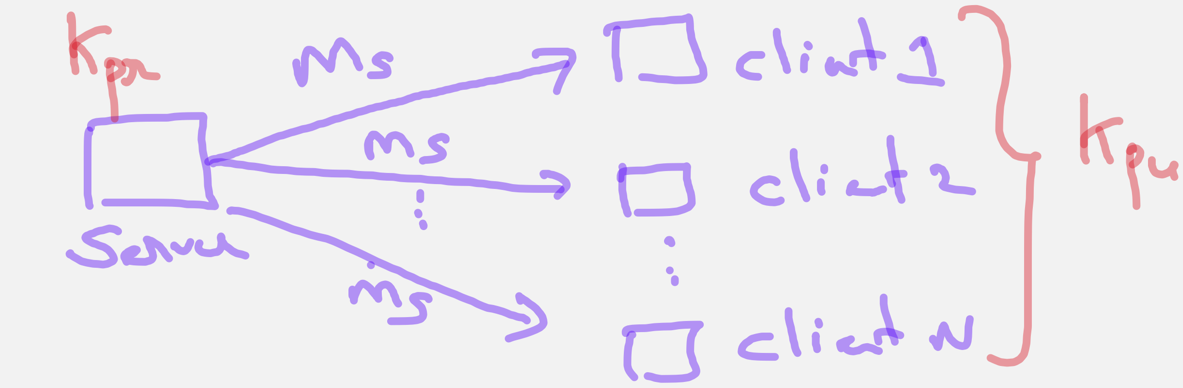
$$\vdots$$

$$D(E(m_N, K_{pu}), K_{pr}) \Rightarrow m_N \ -$$

$$\text{client } 2 \rightarrow \underset{i \neq 2}{\boxed{E(m_i, K_{pu})}} \quad \boxed{K_{pu}}$$

$$\neq \underline{m_i} \quad i \neq 2$$

Authentication

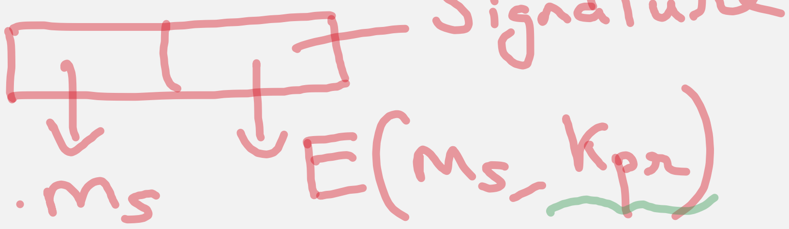

K_pr
Server
m_s
m_s
m_s
client 1
client 2
client N
K_pu

$\{K_{pr}, K_{pu}\}$

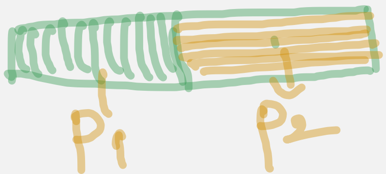
1) Server uses K_{pr} to $\underset{\text{sign}}{\underline{\text{encrypt}}}$ m_s

2) client i uses K_{pu} to verify that m_s is indeed

Coming from the sender



Signature
M_s
$E(M_s, K_{pr})$



$D(P_2, K_{pu})$

```
if $P_1 == D(P_2, K_{pu})$
    verified
else
    something wrong
end
```

Conditions

1) $K_{pu} \not\Rightarrow K_{pr}$!!! } hardness part 1

2) $E(m_i, K_{pu}) \rightarrow$ public } hardness part 2
   $\not\rightarrow$ m_i without having K_{pr}

3) $\underbrace{D(E(m_i, K_{pu}), K_{pr}) = m_i}_{\text{correctness}}$ ✓

RSA

- → Ron Rivest
- → Adi Shamir
- → Leonard M Adleman

1) Key generation idea ✓

2) Encryption is a —

* Hard mathematical problem

prime factorization question?

1) Decide!!! Pick p and q two large prime numbers

$$\boxed{n} = \boxed{pq}$$

2) Reveal n → large number

3) Can you find out its prime factors?

```
for index = 1:1:n
    if index | n
end
```

$p \mid pq$

$p \approx q$

$n \approx p^2$

n is not small

n as an integer } 196 bits

196?

$\{1 \ldots 2$

$\{1, \ldots 2^{128}\}$

```
for index = 1:1:$\sqrt{n}$
    if index | n
end
```

$n \approx 2^{128}$

$\sqrt{n} \approx 2^{64}$

$n \approx 256$ bits $\quad \frac{256}{2}$

exhaustive search $\approx 2^{128}$

$= 2^{128}$

Hard problem 2

1 Dealer → pick integer a

Compute → 3^a

2) Reveal 3^a to public

3) find a ?

$$\log_{10}(3^a)$$

$$\Rightarrow \frac{a\log_{10}(3)}{\log_{10}(3)} = \underline{\underline{\text{"}a\text{"}}}$$

Hard problem

1) Dealer: pick integer
2) pick another integer, b, take b
3) Reveal $\boxed{(b^a) \text{ } \underline{\bmod} \text{ } n.}$

   n is a large number
4) public b and n

   find a?

⊛ prime factorization ←— *

⊛ Discrete logarithm ←— *

"RSA"

a a